

Denoising Carré-du-Champ Optimization on Implicit Data Geometries

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Abstract

Many optimization problems constrain parameters to a data-defined geometry for which neither a parametrization nor an exact projection is available. We study a Jacobian-free alternative based on diffusion geometry. A denoiser supplies an approximate landing map while a positive semidefinite neural carré-du-champ (CDC) field supplies an amortized tangent co-metric. The update applies the CDC directly to the ambient gradient and then lands with the denoiser; it never forms a score Jacobian or inverts a rank-deficient ambient metric. We develop a deterministic inexact-retraction analysis that separates tangent-field error from feasibility error and calibrates the arbitrary CDC scale by a cheap low-rank spectral normalization. The empirical study is intentionally designed around falsifiable diagnostics: projector alignment, normal leakage, feasibility, intrinsic stationarity, objective value, time, and memory. Initial sphere experiments support the scale-normalized construction, whereas torus experiments expose a substantial repeated-denoising drift failure. Thus this draft reports the method and theory target without claiming that the current implementation is solved; the remaining experiments test whether stable landing and uniform geometric accuracy can turn the construction into a practical optimizer.

1 Introduction

Implicit manifolds arise when feasible points are available as samples but a chart, defining equations, and exact projection are unavailable. Examples include learned representations, generated data, diffusion models, and scientific state spaces. Standard Riemannian optimization presumes access to a tangent projector and a retraction. Score-based methods estimate a tangent projector from the Jacobian of a denoiser or score network, but this Jacobian can be expensive and unstable in high ambient dimension. Graph and local-PCA methods avoid neural derivatives but require neighborhood searches at every optimization step.

We propose to learn two objects from noisy samples: a denoiser $R_\theta(y)$ and a PSD low-rank CDC field $G_\phi(y) = M_\phi(y)^\top M_\phi(y)$. The algorithm is

$$x_{k+1} = R_\theta\left(x_k - \eta \hat{G}_\phi(x_k) \nabla f(x_k)\right), \quad \hat{G}_\phi = \frac{G_\phi}{\lambda_{\max}(G_\phi) + \zeta}. \quad (1)$$

The CDC is used as a co-metric/projector, not inverted. The low-rank representation makes both application and spectral calibration independent of a dense $D \times D$ materialization.

Our contributions are: (i) a precise separation of approximate tangent transport and approximate feasibility; (ii) a scale-normalized low-rank CDC update; (iii) a conditional inexact-RGD theorem target with explicit error floors; and (iv) a benchmark protocol that treats feasibility and tangent alignment as first-class outcomes. The current results are preliminary. In particular, a torus failure demonstrates that low denoising MSE alone does not certify a stable retraction.

2 Problem and diffusion geometry

Let $\mathcal{M} \subset \mathbb{R}^D$ be a compact regular embedded manifold and let P_x denote its tangent projector. Given only samples from $\mathcal{M}$, consider minimizing a smooth ambient objective restricted to $\mathcal{M}$:

$$\min_{x \in \mathcal{M}} f(x). \quad (2)$$

For an isometric embedding, the small-time diffusion carré-du-champ matrix of coordinate functions converges, up to convention-dependent scale, to P_x . Consequently $P_x \nabla f(x)$ is the ambient representation of the intrinsic gradient. At finite bandwidth, the population CDC also contains curvature and sampling bias; the learned field has additional estimation and approximation error.

A key implementation point is that G is generally rank deficient. A full inverse is undefined or geometrically meaningless. If $G = M^\top M$ with rank $r \ll D$, then $Gv = M^\top(Mv)$ costs $O(Dr)$ and the largest eigenvalue is the largest eigenvalue of $MM^\top \in \mathbb{R}^{r \times r}$. Normalization in (1) removes the dominant unknown scalar and ensures $\|\hat{G}\|_{\text{op}} \leq 1$ up to ζ .

3 Method

3.1 Conditional learning

For a clean sample X and $Y = X + \sqrt{\varepsilon} \xi$, train R_θ with

$$\mathcal{L}_R = \mathbb{E} \|R_\theta(Y) - X\|^2. \quad (3)$$

The CDC field is trained by a low-rank conditional matching objective. Crucially, the target is centered by the learned conditional mean: $\Delta = X - R_\theta(Y)$

(with the denoiser detached in the CDC update), rather than the uncentered normal displacement $X - Y$. In factor form,

$$\mathcal{L}_G = \mathbb{E} \left[\|M(Y)M(Y)^\top\|_F^2 - \frac{\|M(Y)\Delta\|^2}{\varepsilon} \right], \quad (4)$$

which is the factorized form of matching $G(Y)$ to a conditional covariance target, with constants omitted. The centering removes the conditional normal mean and is important in high ambient dimension. The factorization guarantees positive semidefiniteness and avoids dense matrices.

3.2 Jacobian-free optimization

At each step, automatic differentiation is used only for the scalar objective. The CDC is applied by two matrix-vector products through M_ϕ ; no derivative of R_θ or M_ϕ is formed. The implementation is:

1. compute $g_k = \nabla f(x_k)$ by reverse-mode differentiation of the scalar objective;
2. compute $v_k = \hat{G}_\phi(x_k)g_k$ through the low-rank factor;
3. optionally clip ηv_k to a trust-region radius;
4. set $x_{k+1} = R_\theta(x_k - \eta v_k)$.

For a high-dimensional data set with a strong global linear component, we optionally compute a data-only PCA basis $U \in \mathbb{R}^{D \times m}$ and train/lift the fields in coordinates $U^\top x$; this preprocessing is reported separately from the fully ambient method. We optionally cap the ambient step norm in experiments. This is a practical trust-region safeguard, not part of the basic theorem. When an intrinsic dimension estimate d is available, we also test a Jacobian-free spectral truncation: the top d eigenvectors of $MM^\top$ define the rank- d projector $M^\top Q_d \Lambda_d^{-1} Q_d^\top M$. This separates rank selection from the factor rank and costs an eigendecomposition only in the learned rank dimension.

3.3 Diagnostics and reproducibility

For a known synthetic oracle P , we report $\|\hat{G} - P\|_F$, normal leakage $\|(I - P)\hat{G}\|_F$, $\text{dist}(x, \mathcal{M})$, projected objective $f(\pi(x))$, and $\|P\nabla f(\pi(x))\|$. The objective at an infeasible point is reported separately and is never used as a substitute for the projected objective. Every training run records its seed, bandwidth schedule, rank, step cap, training losses, and per-method runtime in JSON. Large checkpoints and generated figures are ignored by git, while code, configs, logs, and the manuscript are versioned.

4 Inexact convergence analysis

We state the deterministic result in a form that makes its assumptions auditable. Let π be the closest-point map on a tube of radius below the reach. Assume the iterates and all trial points remain in that tube, f has L -Lipschitz gradient and bounded gradient C , and

$$\|R(y) - \pi(y)\| \leq \delta_R, \quad \|G(x) - P_{\pi(x)}\| \leq \delta_G. \quad (5)$$

The first bound is a landing error and the second is a tangent co-metric error; neither is implied by training loss without a generalization argument.

Theorem 1 (Tube-stable inexact CDC-RGD). *Let $\mathcal{M}$ have reach at least τ , let π be its closest-point map on $\mathcal{T}_{\tau_0}(\mathcal{M})$ with $\tau_0 < \tau$, and let f have L -Lipschitz ambient gradient and gradient bound C on this tube. Suppose R and G satisfy (5), $\|G(x)\|_{\text{op}} \leq 1$, and the learned update is well-defined whenever $\text{dist}(x, \mathcal{M}) \leq \delta_R$ and $\eta C + \delta_R < \tau_0$. Assume the exact projected step satisfies*

$$f(\pi(p - \eta P_p \nabla f(p))) \leq f(p) - a\eta \|P_p \nabla f(p)\|^2 \quad (6)$$

for all $p \in \mathcal{M}$ and some $a > 0$. Then, from any $x_0 \in \mathcal{M}$, all iterates satisfy $\text{dist}(x_k, \mathcal{M}) \leq \delta_R$, and with $p_k = \pi(x_k)$,

$$\frac{1}{N} \sum_{k=0}^{N-1} \|\text{grad}_{\mathcal{M}} f(p_k)\|^2 \leq \frac{f(p_0) - f_*}{a\eta N} + \frac{C_f L_\pi}{a\eta} [(1 + L_\pi)\delta_R + L_\pi \eta (C\delta_G + L\delta_R)], \quad (7)$$

where C_f is a Lipschitz constant for f on the tube and L_π is a Lipschitz constant for π .

Proof. Let $T(p) = \pi(p - \eta P_p \nabla f(p))$. Since $\|G\|_{\text{op}} \leq 1$, the trial point is in the tube by the step-size assumption. Comparing the learned trial point with the exact trial point at p_k gives

$$\|x_k - \eta G(x_k) \nabla f(x_k) - (p_k - \eta P_{p_k} \nabla f(p_k))\| \leq \delta_R + \eta(C\delta_G + L\delta_R).$$

The landing error and Lipschitzness of π imply

$$\|x_{k+1} - T(p_k)\| \leq \delta_R + L_\pi[\delta_R + \eta(C\delta_G + L\delta_R)] =: \rho.$$

Applying π once more gives $\|p_{k+1} - T(p_k)\| \leq L_\pi \rho$. Lipschitzness of f , (6), and telescoping over k yield (7); feasibility follows directly from $\|R(y) - \pi(y)\| \leq \delta_R$. $\square$

For completeness, the exact descent assumption follows from the reach. If the closest-point retraction obeys the standard second-order estimate $\|\pi(p + \xi) - p - \xi\| \leq \kappa \|\xi\|^2$ for tangent ξ , then with $\xi = -\eta P_p \nabla f(p)$, smoothness gives

$$f(T(p)) \leq f(p) - \eta \|\text{grad}_{\mathcal{M}} f(p)\|^2 + B\eta^2 \|\text{grad}_{\mathcal{M}} f(p)\|^2,$$

where $B = C\kappa + \frac{L}{2}(1 + \kappa\eta C)^2$. Thus $\eta \leq (2B)^{-1}$ gives (6) with $a = 1/2$. The theorem exposes the unavoidable deterministic floor from a learned landing map: at fixed η , δ_R contributes as δ_R/η , while tangent-field error contributes as δ_G . A landing-flow variant requires a separate normal contraction proof.

The statistical interpretation is

$$\delta_R \leq \delta_{R,\text{approx}} + \delta_{R,\text{stat}}(n, \varepsilon) + \delta_{R,\text{bias}}(\varepsilon), \quad (8)$$

$$\delta_G \leq \delta_{G,\text{approx}} + \delta_{G,\text{stat}}(n, \varepsilon) + \delta_{G,\text{bias}}(\varepsilon). \quad (9)$$

The optimization theorem is conditional on these uniform errors; it does not assert that CDC matching automatically achieves them.

5 Experimental protocol

The current implementation evaluates synthetic manifolds with known projection and tangent projector only for evaluation: spheres, embedded tori, Swiss rolls, and disconnected components. Variable-density and branched supports are planned next; at a branch there is no single smooth tangent oracle, so those experiments require set-valued or diffusion-geometric metrics. Objectives are random smooth quadratics with held-out start points and independent random seeds. Baselines are oracle projected RGD, score-Jacobian RGD, graph/local-PCA RGD, latent pullback optimization where a chart is available, Euclidean penalty, and denoiser-only landing.

The intended reporting protocol includes final and projected objective, distance to the manifold, tangent projector error, normal leakage, intrinsic gradient norm, wall-clock, peak memory, and failure rate. The current CPU prototype records all of these except failure-rate aggregates; peak resident memory is recorded per run (about 1.01 GB in the five-seed torus study). Bandwidth, rank, sample count, denoiser quality, step size, and objective smoothness are swept. High-dimensional experiments test whether the $O(Dr)$ CDC application avoids the memory growth of score Jacobians.

6 Initial results and limitations

The current implementation is a diagnostic prototype in PyTorch. On a sphere in $\mathbb{R}^{16}$, spectral normalization and identity initialization of the residual denoiser kept CDC-RGD feasible to about 0.014 mean distance and reached projected objective 10.85, compared with oracle objective 10.48. These numbers are single-seed and not a superiority claim.

An early torus run in $\mathbb{R}^{16}$ exhibited distance around 0.7 after repeated steps, despite low training denoising loss. Identity initialization of the residual denoiser plus a step cap reduced this in a five-seed sweep to 0.024 ± 0.007 , but CDC-RGD still reached mean projected objective 11.93 versus 8.73 for oracle RGD, 9.30 for the penalty baseline, and 10.84 for score-Jacobian RGD. With the corrected centered CDC target, the equal-budget held-out diagnostics averaged

normalized-field projector error 0.190 ± 0.009 and normal leakage 0.080 ± 0.006 . Thus the stability safeguard improves feasibility but does not eliminate geometry error. Smaller bandwidth increased error in an earlier study. These results falsify the naive claim that denoiser MSE is a sufficient retraction certificate and that feasibility alone implies useful optimization. A staged bandwidth schedule ($0.1 \rightarrow 0.03 \rightarrow 0.01$) produced the corrected values above. With a 0.03 step cap and 100 steps, CDC-RGD reached projected objective 10.16 versus 9.81 for oracle RGD and 10.56 for the penalty baseline, with distance 0.012; the score-Jacobian baseline reached 12.15. The comparison is single-seed and synthetic, but indicates that the earlier CDC deficit was partly a too-conservative step cap. The local-PCA baseline (30 baseline steps in this prototype) reached projected objective 12.16 at 3.62 s, versus CDC-RGD 10.16 at 0.48 s; score-Jacobian (also 30 steps) reached 12.15 at 3.07 s, and penalty was 10.56 at 0.34 s. Runtime/objective matching across equal iteration budgets remains to be done.

The subsequent five-seed held-out equal-budget torus continuation study is summarized in Table 1. CDC is competitive with all non-oracle baselines while preserving feasibility and has much lower inference time than Jacobian/local-PCA methods, but remains behind the oracle. A separate single-seed prototype run showed the same ordering (projected objectives 10.16 for CDC, 10.51 score-Jacobian, 10.54 local-PCA, 10.56 penalty, and 9.81 oracle). Under the earlier uncentered CDC target, a first ambient- $D = 256$ sphere run using the low-dimensional scalar bandwidth failed (CDC distance 2.47). Dimension-scaled noise improved the denoiser clean-point error to 0.010, but the learned CDC remained unusable (CDC distance 0.90); the associated raw diagnostic also showed severe tangent leakage. The fully ambient high-dimensional method needs a different estimator or preconditioning: an ambient-64 sphere run failed (CDC distance 0.35, normalized projector error 1.00, normal leakage 0.99), a rank-4/projector-rank-3 ambient-256 rerun failed, and increasing an ambient-64 centered training set to 100k samples still gave CDC distance 0.183 and projector error 0.972. However, a data-only PCA preconditioner (eight retained components) followed by the lifted centered CDC model succeeded on the ambient-256 sphere: CDC reached projected objective 140.873 versus oracle 140.872, distance 0.002, and projector error 0.261. This is an important qualified scaling result: global linear compression can rescue a nearly linear high-dimensional support, but is not a substitute for a general nonlinear high-dimensional estimator. As a separate disconnected-support smoke test, two separated spheres gave CDC projected objective 17.26 and distance 0.016, versus penalty objective 17.34 and distance 0.051; this is not yet a multi-seed result. A three-seed robustness sweep with the centered target is informative. With Gaussian training noise of standard deviation 0.05, CDC reached projected objective 8.11 ± 3.14 and distance 0.264 ± 0.399 , versus penalty 7.82 and distance 0.437, score-Jacobian 9.39 and distance 0.148, and local-PCA 9.50 and distance 0.063; projector error increased to 0.580 and normal leakage to 0.542. Under angular density concentration 0.4, CDC reached 8.09 ± 2.95 and distance 0.021 ± 0.003 , versus penalty 7.85 and distance 0.070, score-Jacobian 8.75, and local-PCA 8.76; projector error was

Table 1: Five-seed held-out equal-100-step torus continuation study (mean $\pm$ standard deviation). Lower is better for all columns.

method	projected f	distance	intrinsic $\ g\ $	runtime (s)
CDC-RGD	9.08 ± 4.49	$0.013 \pm .002$	$0.80 \pm .14$	0.316
penalty	9.26 ± 4.43	$0.043 \pm .011$	$0.84 \pm .13$	0.345
score-Jacobian	9.49 ± 4.50	$0.013 \pm .002$	$1.48 \pm .34$	5.191
local-PCA	9.49 ± 4.50	$0.013 \pm .002$	$1.46 \pm .36$	5.991
oracle RGD	8.74 ± 4.55	$< 10^{-7}$	$0.12 \pm .09$	0.194

0.165 and leakage 0.069. The clean control gave CDC 7.81, penalty 7.91, and score/local-PCA 9.35, with distance 0.016. These centered results support density robustness but expose substantial noise sensitivity. Conversely, a corrected equal-100-step three-seed run with only 10k training samples was unstable: CDC projected objective 8.86 ± 2.44 and distance 0.334 ± 0.441 , while penalty achieved 7.55 and distance 0.506, score-Jacobian 8.45 and distance 0.304, and oracle 7.27. This sample-size sensitivity rules out a broad equal-budget superiority claim. Repeating the equal-100-step study with 20k samples (three seeds) stabilized the method: CDC projected objective 7.97 ± 2.30 and distance 0.014 ± 0.002 , penalty 7.95 ± 2.82 and distance 0.051 ± 0.022 , score-Jacobian 8.60 ± 2.01 and distance 0.015 ± 0.002 , local-PCA 8.61 ± 2.00 and distance 0.015 ± 0.002 , and oracle 7.44 ± 2.89 . CDC runtime was 0.316 s versus 5.19 s for score-Jacobian and 5.99 s for local-PCA. Paired across the five seeds, centered CDC beat penalty by 0.183 ± 0.103 objective units and had 0.030 ± 0.013 lower distance, winning all five seeds; it beat score-Jacobian and local-PCA by 0.407 ± 0.084 and 0.414 ± 0.069 objective units, respectively, also winning all five. These paired results are encouraging but remain limited to one synthetic family and random quadratic objectives, not a statistically powered generalization claim. A variable-density torus smoke test (angular concentration 0.4) gave CDC projected objective 10.52 and distance 0.014, versus penalty 10.80 and distance 0.026, score-Jacobian 10.86 and distance 0.014, and local-PCA 10.87 and distance 0.015; normalized projector error was 0.225 and normal leakage 0.142. A controlled density sweep remains future work.

7 Related work

This work connects diffusion geometry, score/denoiser retractions, metric matching, and inexact Riemannian optimization. Recent diffusion-geometry methods motivate learning tangent information from noisy samples and provide finite-bandwidth/finite-sample error decompositions [4, 3]; score-based landing methods instead differentiate a denoiser or score [5, 1]. Metric matching provides a scalable route to learning PSD geometric fields [2]. Inexact Riemannian optimization provides a relevant convergence framework [6], but the present setting must separately control a learned co-metric and a learned landing map on an

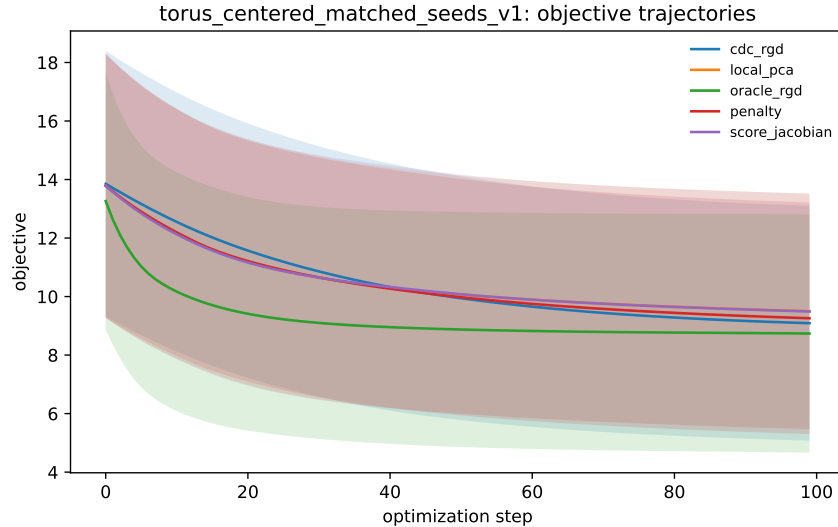


Figure 1: Held-out torus objective trajectories over five equal-budget seeds. Shaded regions show one standard deviation; all methods use 100 optimization steps.

unknown embedded support.

8 Conclusion

CDC-RGD is a promising interface between diffusion geometry and optimization, but the central scientific question is stability rather than a low objective on one run. The current evidence supports spectral calibration and exposes a concrete torus failure. A complete NeurIPS result requires proving the off-manifold perturbation theorem in full, establishing estimator rates or empirical uniform certificates, and completing the controlled benchmark suite.

A Implementation details

All neural fields use three hidden SiLU layers and AdamW with learning rate 2×10^{-3} and weight decay 10^{-5} . The final denoiser layer is initialized to zero, so the residual denoiser starts at the identity. The factor field uses fresh Gaussian perturbations on each minibatch and is trained with the current denoiser as a detached conditional-mean center. Unless otherwise stated, the continuation schedule uses 150 total epochs staged equally across $\varepsilon = 0.1, 0.03, 0.01$ (the smaller smoke studies use 100 total epochs), batch size 512, hidden width 256, factor rank 8, and spectral projector rank 2. The optimization objective is a

random positive-definite quadratic plus a linear term; all methods in a run use the same objective and held-out starts.

The oracle baseline uses the analytic closest-point projection and tangent projector of the synthetic manifold. The penalty baseline differentiates the denoiser-based objective plus a landing penalty of 2 and uses one quarter of the CDC nominal step size. The score-Jacobian baseline differentiates the denoiser output with respect to each input sample; local-PCA uses 64 nearest training samples and the leading two covariance eigenvectors. Equal-budget experiments set all methods to 100 optimization steps; the earlier continuation table deliberately labels its 30-step score/local-PCA baseline separately.

B Proof details for the tube theorem

We record the perturbation calculation used in the theorem. Write $p = \pi(x)$ and $g_p = P_p \nabla f(p)$. Since $\|x - p\| \leq \delta_R$, $\|G - P_p\| \leq \delta_G$, $\|G\|_{\text{op}} \leq 1$, and ∇f is L -Lipschitz,

$$\begin{aligned} \|G(x) \nabla f(x) - g_p\| &\leq \|(G(x) - P_p) \nabla f(x)\| + \|G(x) (\nabla f(x) - \nabla f(p))\| \\ &\leq C\delta_G + L\delta_R. \end{aligned}$$

Thus the learned trial point differs from the exact trial point by at most $\delta_R + \eta(C\delta_G + L\delta_R)$. Applying the landing error and the Lipschitz closest-point map gives the ρ bound in the theorem. Applying π to the learned endpoint compares p_{k+1} to the exact endpoint $T(p_k)$, and Lipschitzness of f converts this endpoint perturbation into an additive descent error. Summing the resulting inequality over k telescopes only the projected values $f(p_k)$, avoiding an erroneous accumulation of the off-manifold objective error. This proof also explains why a fixed landing error produces a δ_R/η stationarity floor.

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